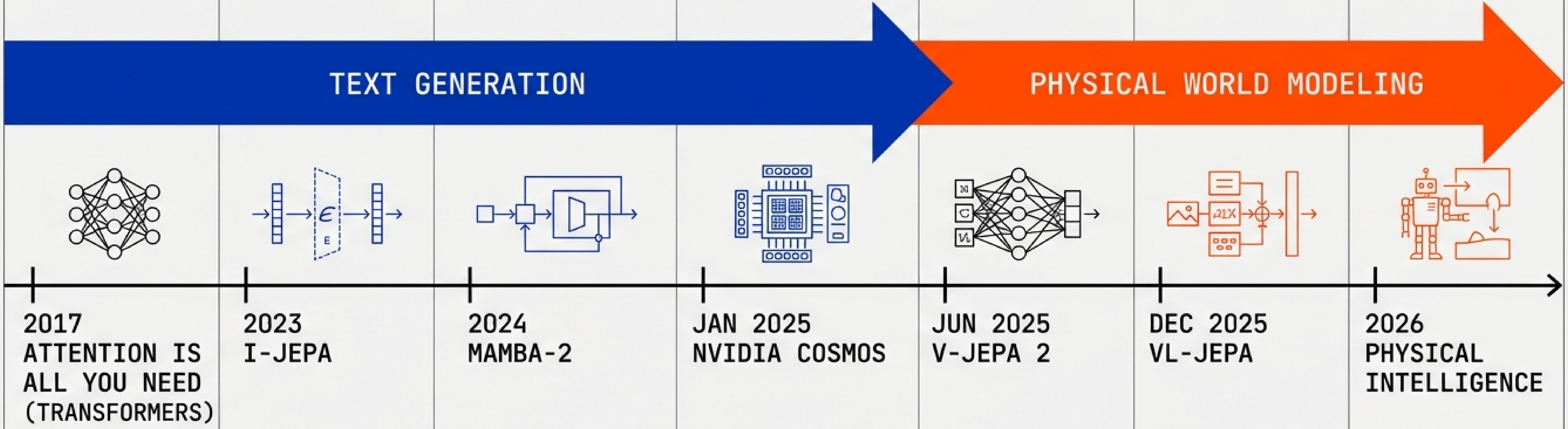
THE EVOLUTION OF MODERN AI ARCHITECTURES

FROM TRANSFORMERS TO PHYSICAL WORLD MODELS: A TECHNICAL DEEP DIVE

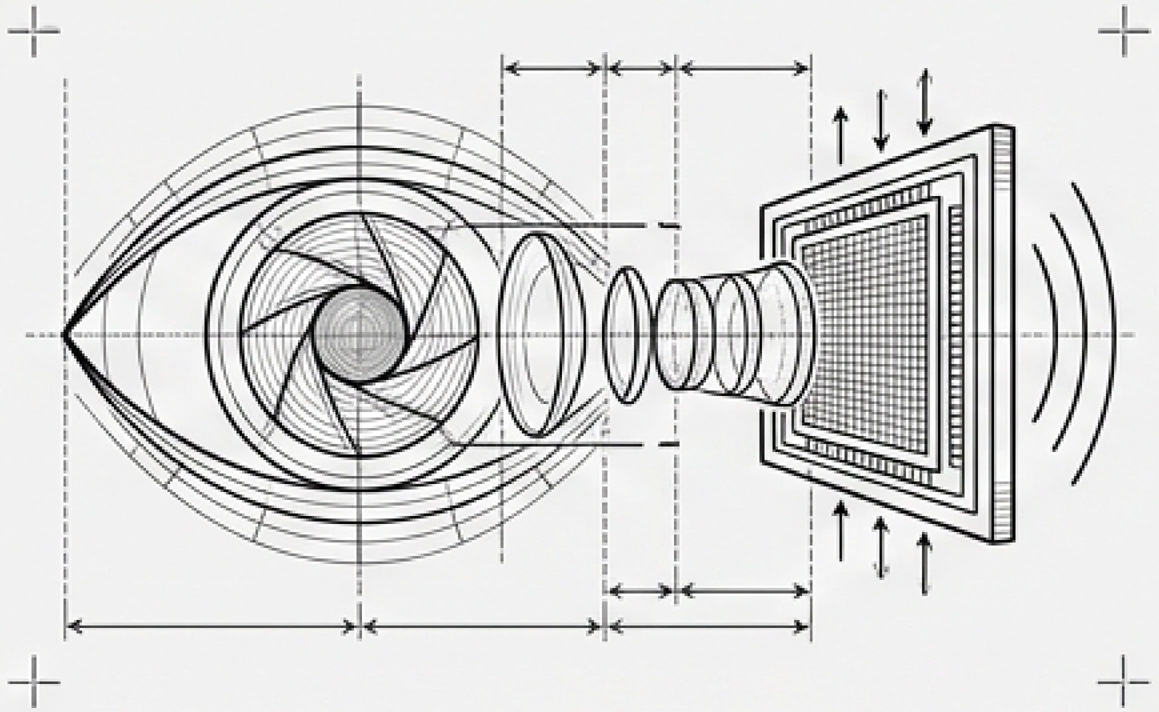
SCOPE: MAMBA-3 / VL-JEPA / VLA / NVIDIA COSMOS | AUDIENCE: AI ENGINEERS & RESEARCHERS | TIMELINE: 2026 PRODUCTION REALITY



2026 PRODUCTION REALITY: THE ARCHITECTURE DECISION TREE

OPTIMIZING FOR CONSTRAINT vs. SCALE

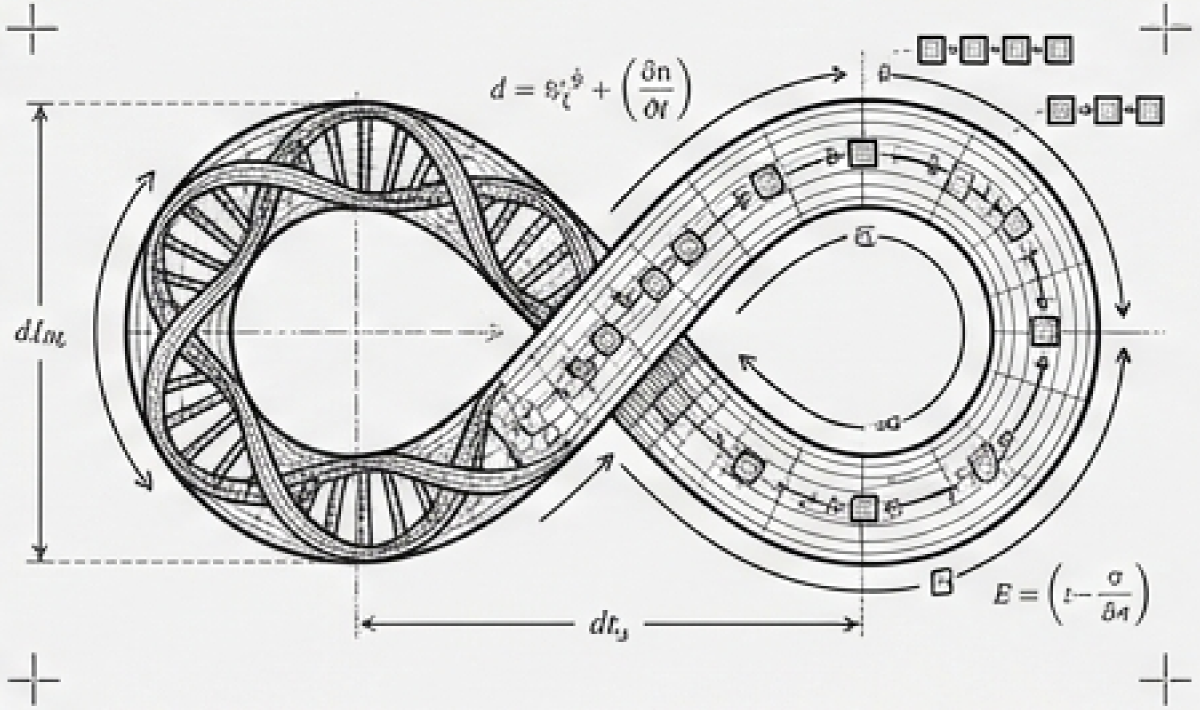
CONSTRAINT: REAL-TIME PERCEPTION / EDGE



SOLUTION: VL-JEPA

Semantic embedding prediction.
Single forward pass.
2.85x fewer operations on streaming video.

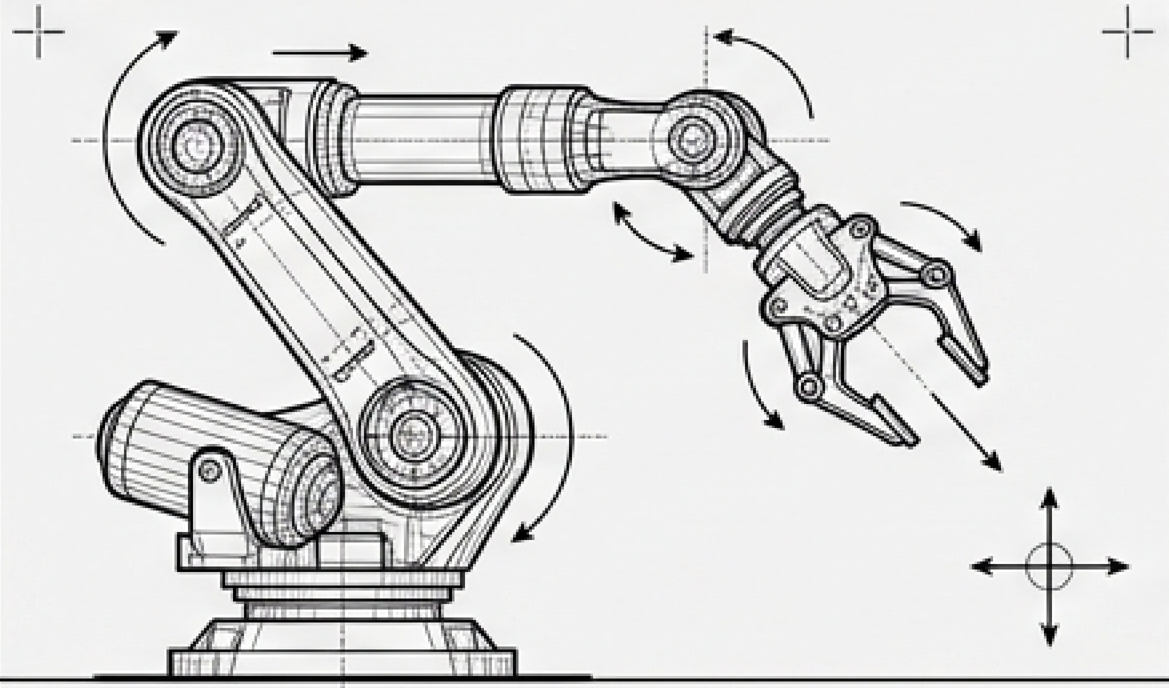
CONSTRAINT: LONG-CONTEXT REASONING (>100K TOKENS)



SOLUTION: MAMBA-3

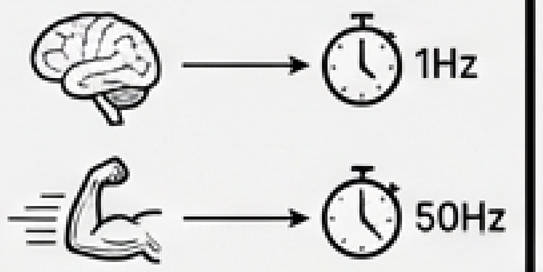
Linear O(n) state-space complexity. 5x faster than Transformers.
Solves state tracking.

CONSTRAINT: PHYSICAL CONTROL & ROBOTICS

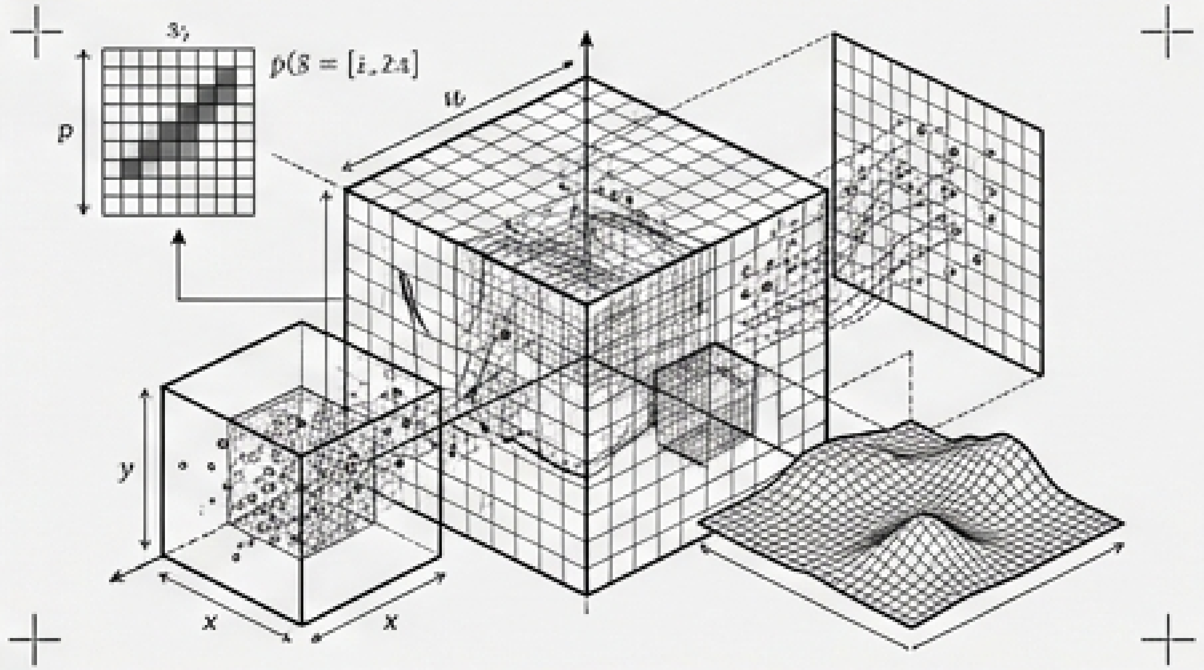


SOLUTION: DUAL-SYSTEM VLA

Decoupled architecture.
System 2 (Reasoning) @ 1Hz.
System 1 (Actuation) @ 50Hz.

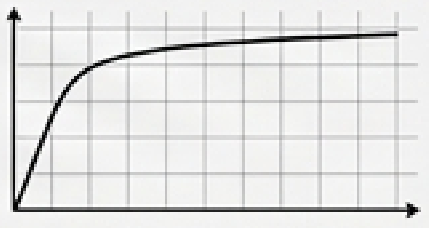


CONSTRAINT: HIGH-FIDELITY SIMULATION



SOLUTION: NVIDIA COSMOS / DIFFUSION

Pixel-perfect rollout.
Trained on 20M+ hours of physical data.



THE TRANSFORMER STANDARD & THE QUADRATIC BOTTLENECK

WHY ATTENTION FAILS IN PHYSICAL DOMAINS

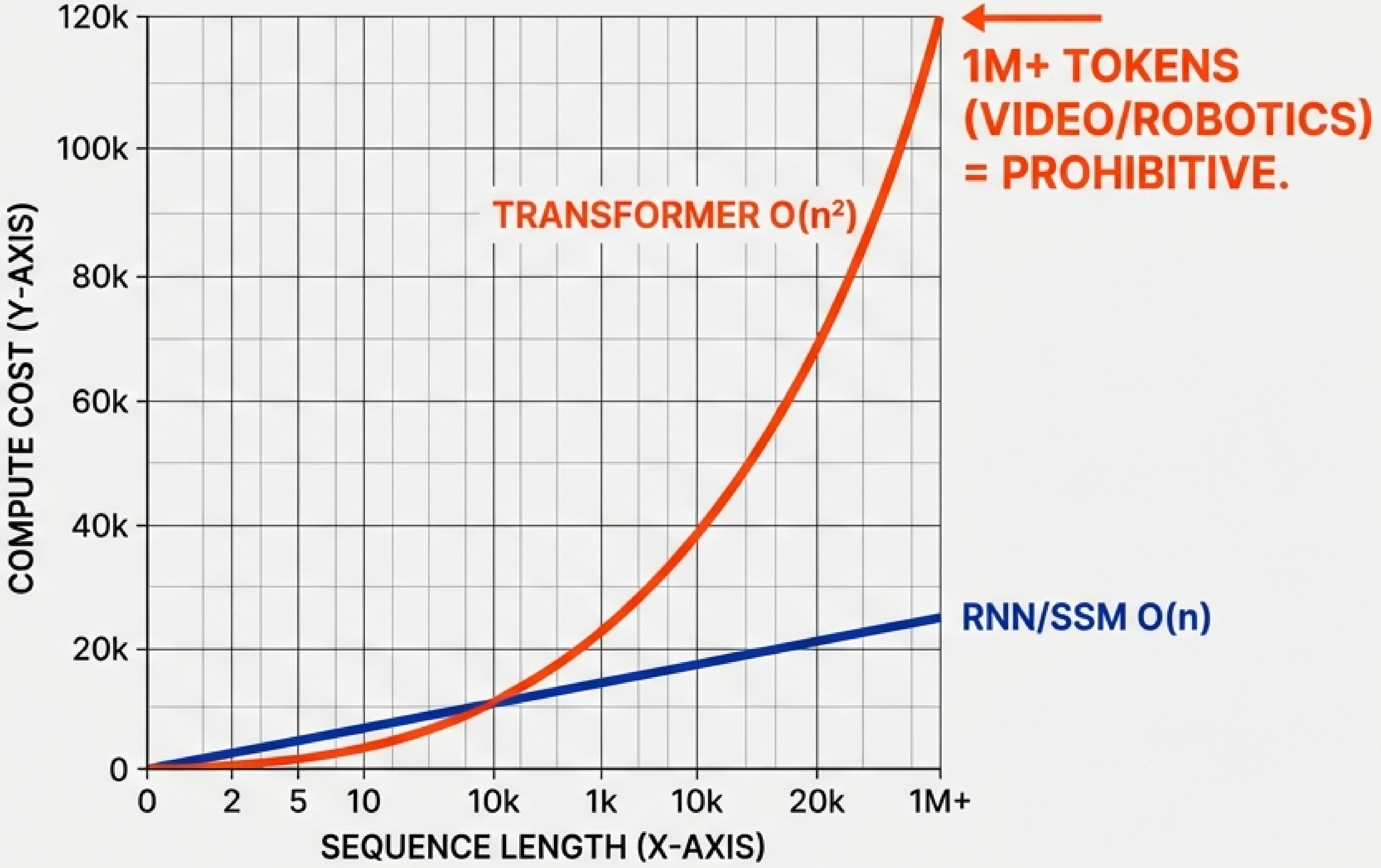
THE INCUMBENT KING

- **Mechanism:** Self-Attention weights token significance relative to every other token.
- **Dominance:** Unmatched expressivity in text/code (GPT-4, Llama).
- **Training:** Highly parallelizable.

THE PHYSICS GAP

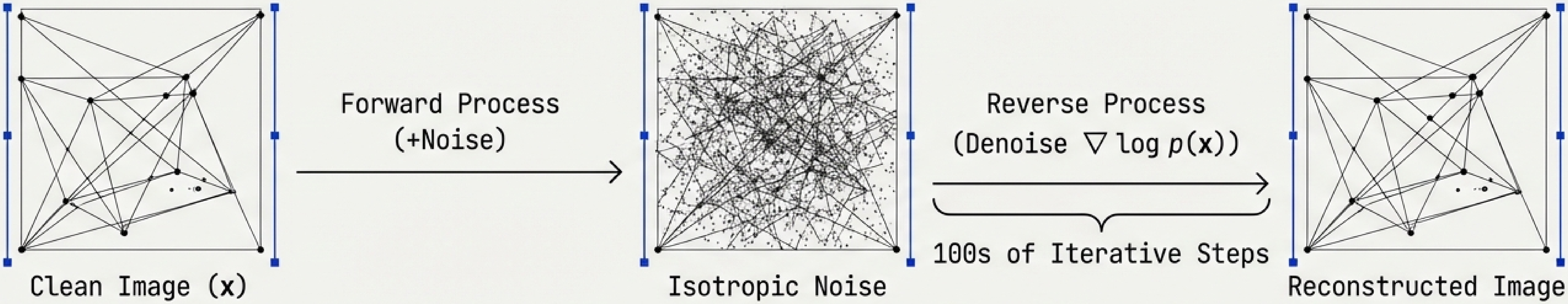
- Transformers model correlations, not causal physics.
- **Failure Mode: Hallucination in text = Crash in robotics.**

COMPUTE SCALING VISUALIZATION



DIFFUSION MODELS: HIGH FIDELITY, HIGH LATENCY

THE PIXEL-PERFECT TRAP



STATE-OF-THE-ART TRAINING

Technique: Flow Matching & Rectified Flow.

Optimization: Straighter trajectory modeling over standard DDPM.

THE REAL-TIME LIMITATION

Inference Penalty: Iterative sampling is too slow for 50Hz control loops.

Efficiency Gap: Models every texture, shadow, and artifact.

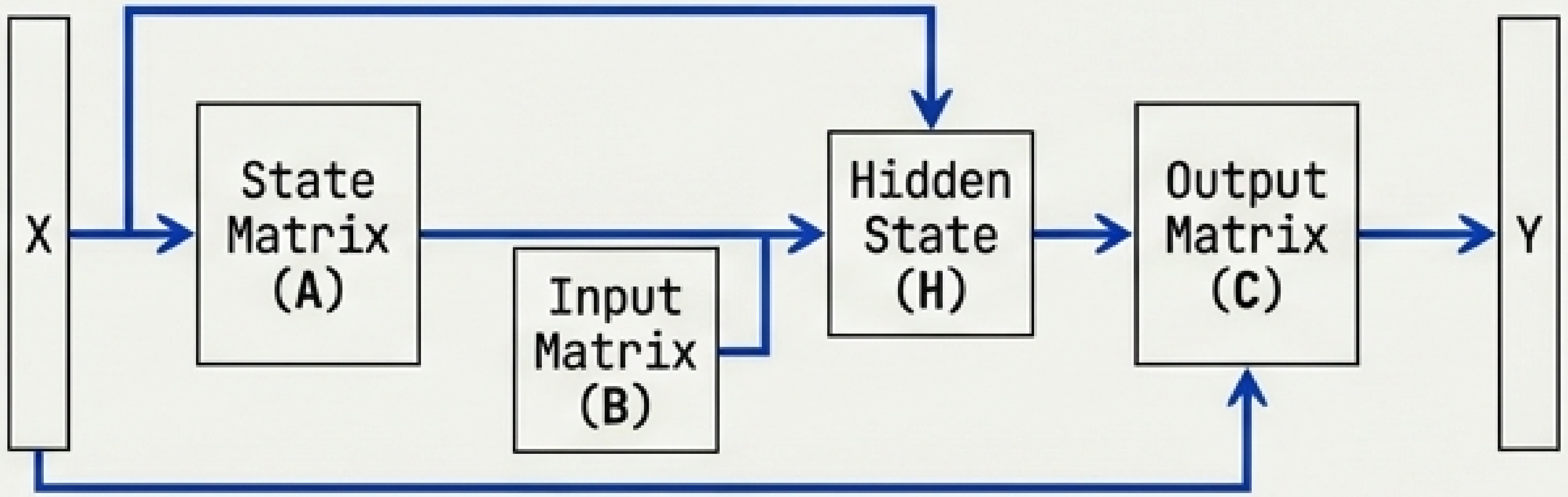
Waste: Computationally expensive for agents needing only semantic ‘obstacle’ detection.

MAMBA-3: THE INFERENCE-FIRST SSM

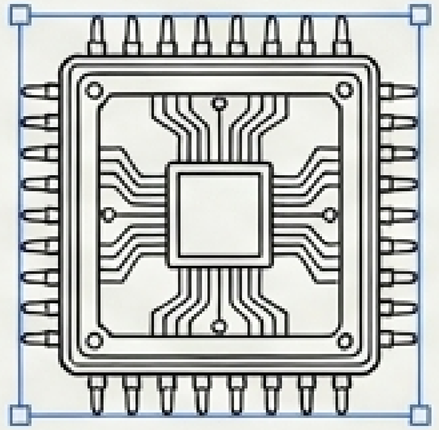
SOLVING THE MEMORY BOTTLENECK

CORE CONCEPT: SELECTIVE STATE SPACE MODEL

- **Complexity:** Linear O(n) time.
- **Memory:** Constant O(1) during generation.
- **Structure:** Compresses context into a fixed-size state.



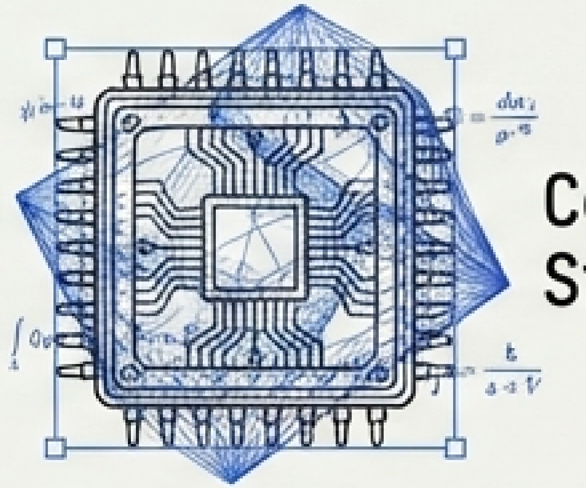
EVOLUTIONARY LEAP



Tensor Core Optimized (SSD)

Fails Parity/Copy Tasks

Mamba-2: Optimized for speed but struggled with precise recall.



Complex State

100% Accuracy on State Tracking

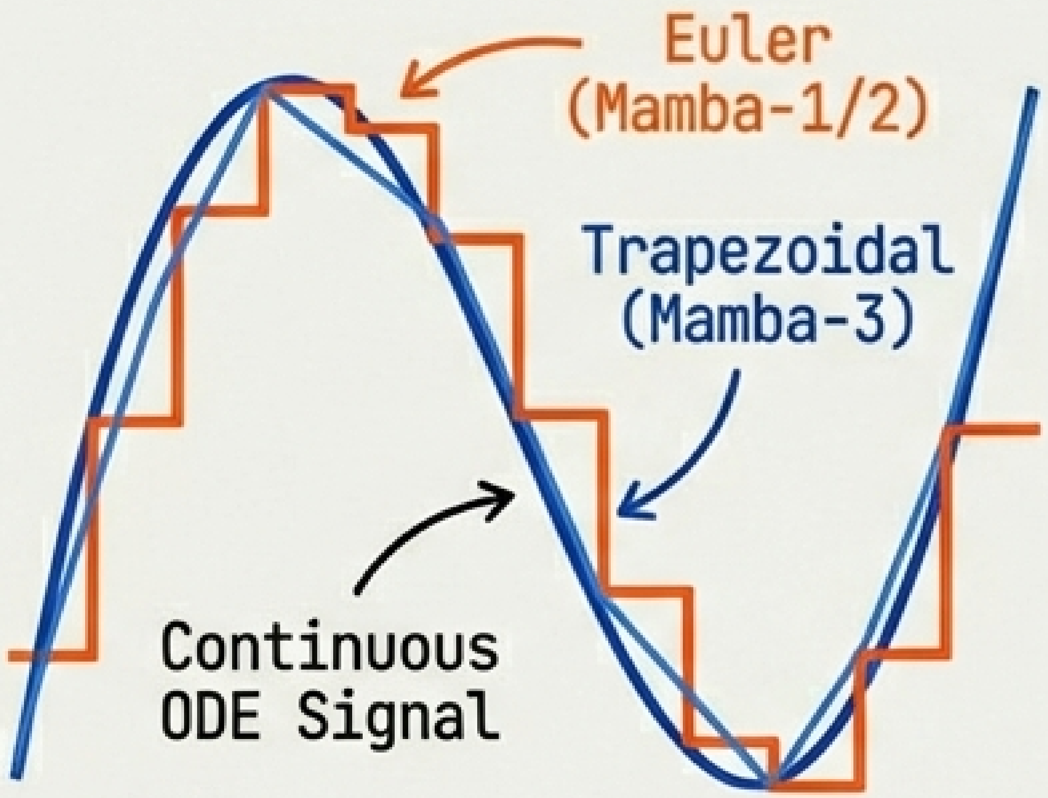
Mamba-3: Retains hardware efficiency, solves 'forgetting' and oscillation.

NotebookLM

MAMBA-3 ARCHITECTURE: DISCRETIZATION & STATE UPDATES

MATHEMATICAL INNOVATIONS FOR STABILITY

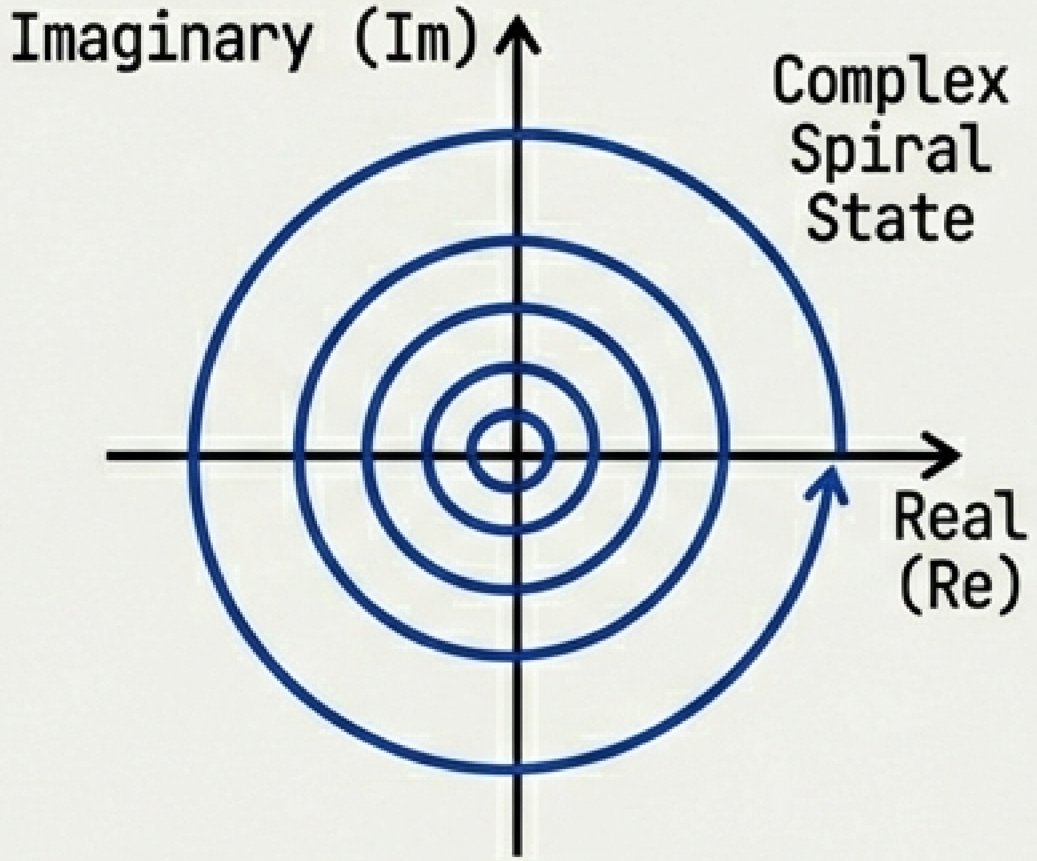
TRAPEZOIDAL DISCRETIZATION



Higher fidelity approximation of continuous.

Improves long-context stability.

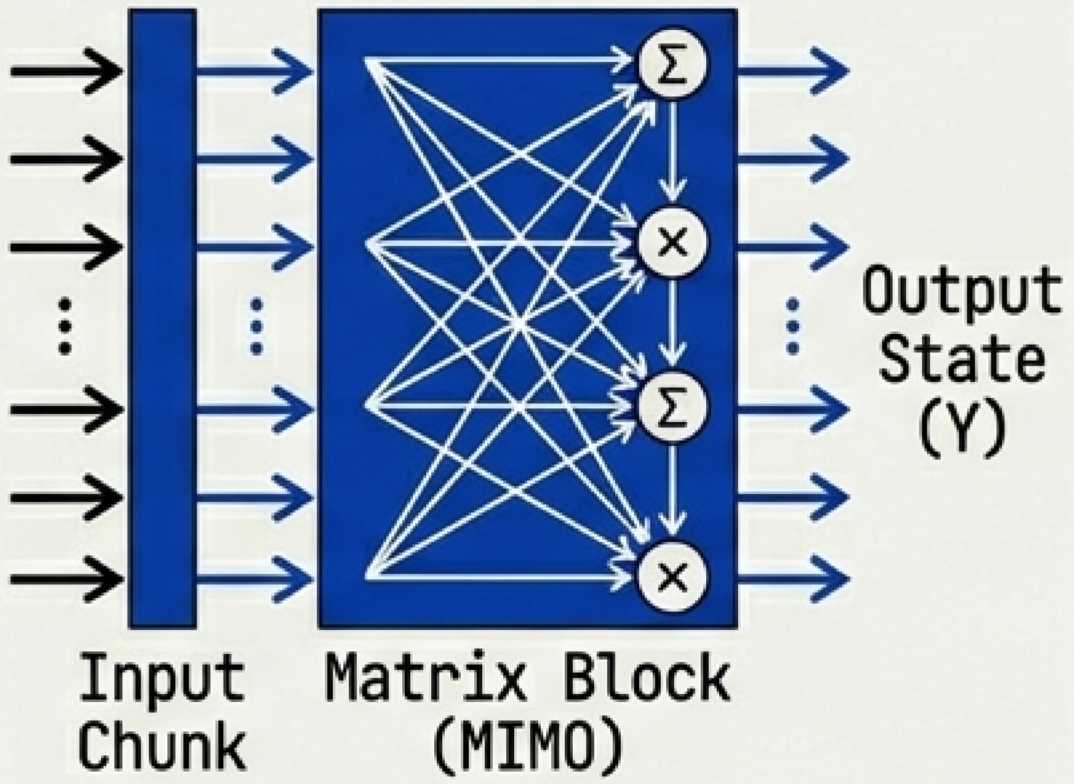
COMPLEX-VALUED STATE



Introduction of complex numbers in transition matrices.

Naturally models oscillatory dynamics and parity tasks where real-valued SSMs fail.

MIMO FORMULATION



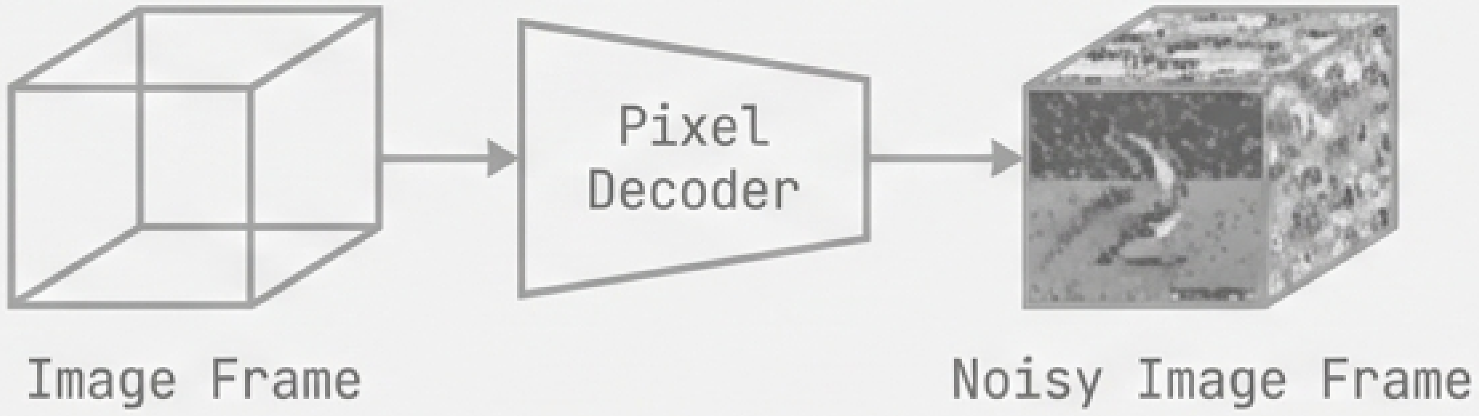
Multi-Input Multi-Output.

Processes inputs in chunks to saturate Tensor Core arithmetic intensity.

VL-JEPA: SHIFTING FROM GENERATION TO PREDICTION

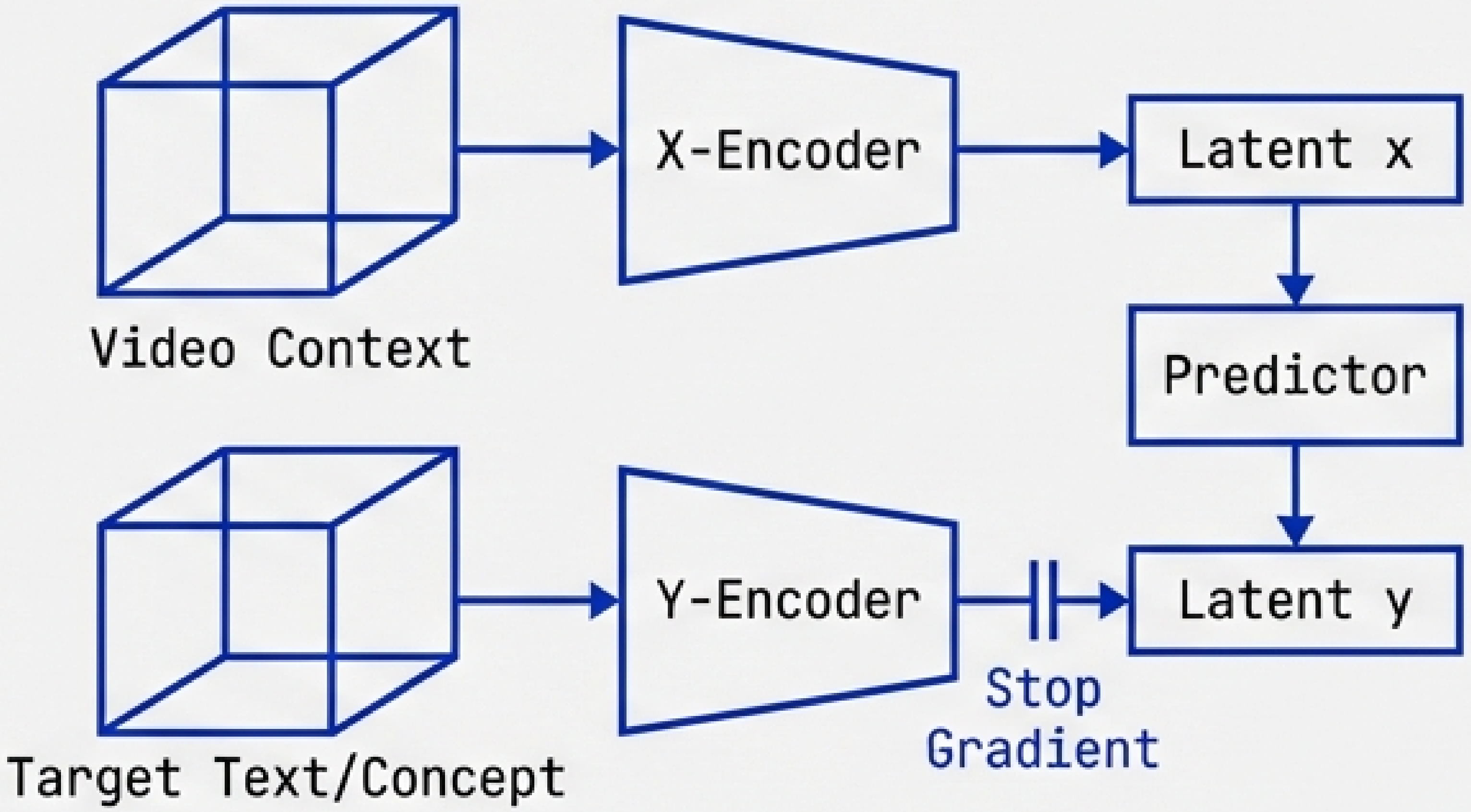
PREDICTING EMBEDDINGS, NOT PIXELS

GENERATIVE PARADIGM (Diffusion/Transformer)



Predicts x(t+1) (Pixels).
Costly. Noise-sensitive.

PREDICTIVE PARADIGM (JEPA)

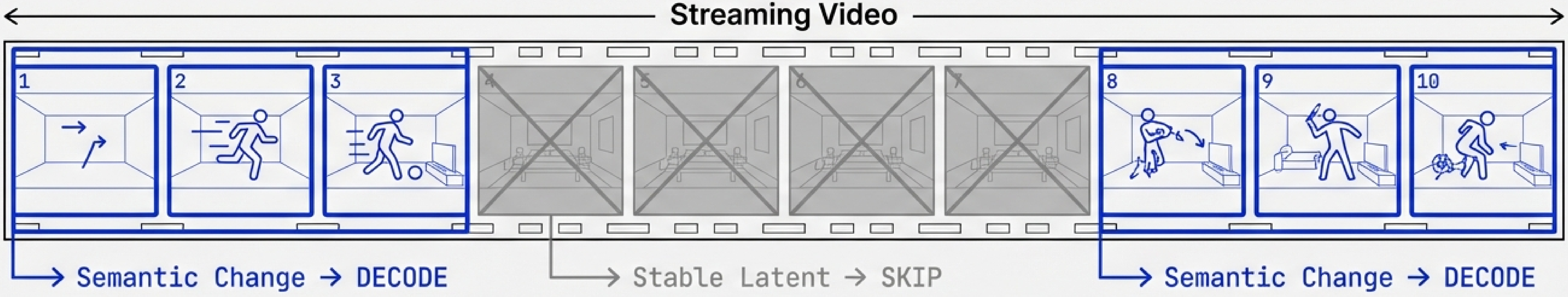


Predicts y(t+1) (Abstract Embeddings).
Efficient. Semantic-focused.

Loss Function: InfoNCE (Contrastive).
No pixel reconstruction.

VL-JEPA PERFORMANCE: SPEED & SELECTIVE DECODING

EFFICIENCY AT THE EDGE



INFERENCE EFFICIENCY
2.85x fewer decoding operations vs standard VLMs.
Single forward pass. No autoregressive wait times.

PARAMETER EFFICIENCY
1.6B Parameters.
Outperforms GPT-4o on 'Perception Test' & 'TempCompass'.
50% fewer parameters than pixel-based peers.

NVIDIA COSMOS: THE WORLD FOUNDATION MODEL SIMULATING REALITY FOR PHYSICAL AI

DEFINITION

A World Foundation Model (WFM) designed to simulate physical interactions for robotics and AVs.

CORE PHILOSOPHY

High-fidelity physics-based video generation.

The "Training Ground" for embodied agents.

TRAINING SCALE

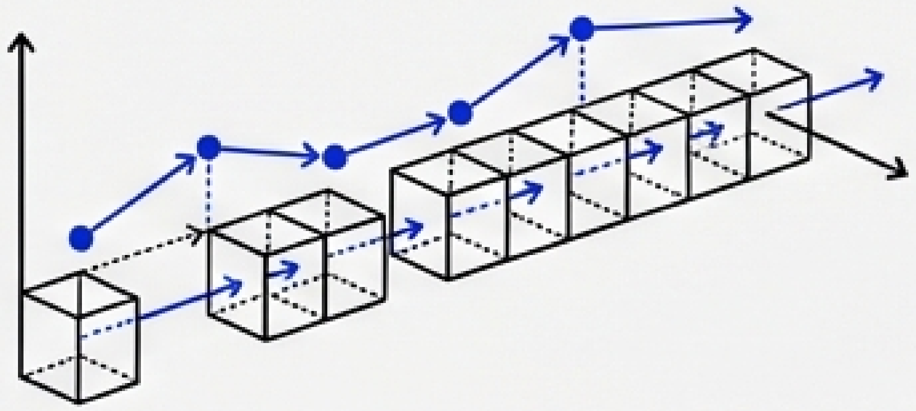
20,000,000 HOURS

of Video Data (Driving, Robotics, Human Interaction).

9,000 Trillion Tokens.

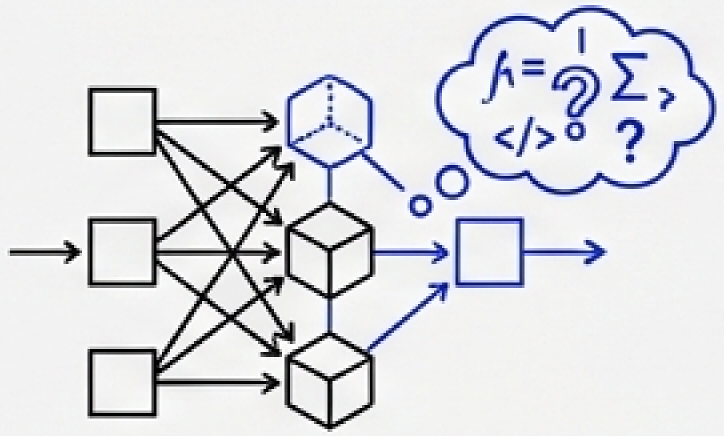
THE COSMOS TRIAD ARCHITECTURE

PREDICT, REASON, TRANSFER



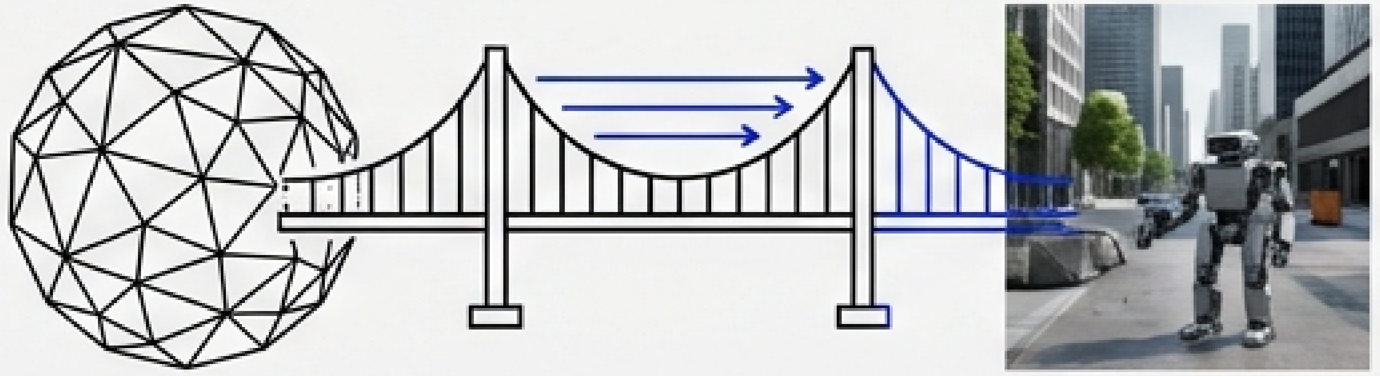
COSMOS PREDICT (FUTURE STATE)

Generates multi-frame rollouts (up to 30s).
Predicts intermediate actions and motion trajectories.



COSMOS REASON (THE BRAIN)

7B Parameter VLM.
“Chain-of-Thought” reasoning about physical outcomes (e.g., glass shattering physics).



COSMOS TRANSFER (DOMAIN ADAPTATION)

Sim-to-Real Bridge.
Generates photorealistic synthetic data.
3.5x smaller/faster than previous generation.

VLA (VISION-LANGUAGE-ACTION): TOKENIZING PHYSICS

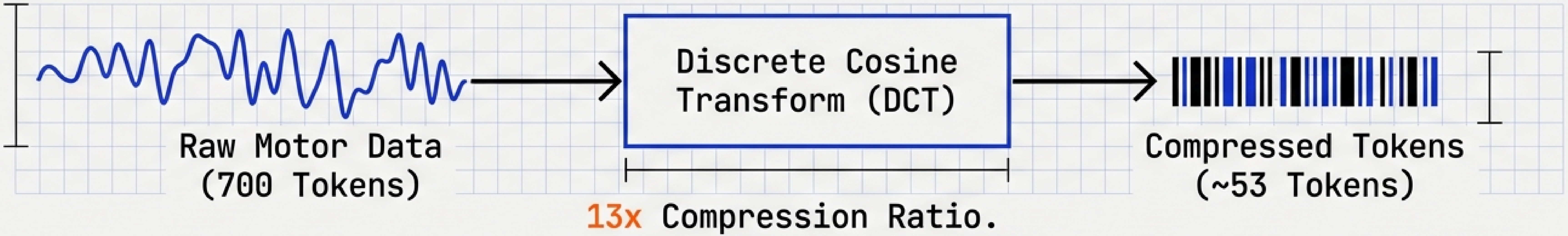
BRIDGING THE DISCRETE-CONTINUOUS GAP

OUTPUT MODALITY SHIFT

VLM Output = Text <token> 'Open the door'

VLA Output = Action <act> Δx, Δy, Δz, gripper_state

FAST TOKENIZATION VIA DCT

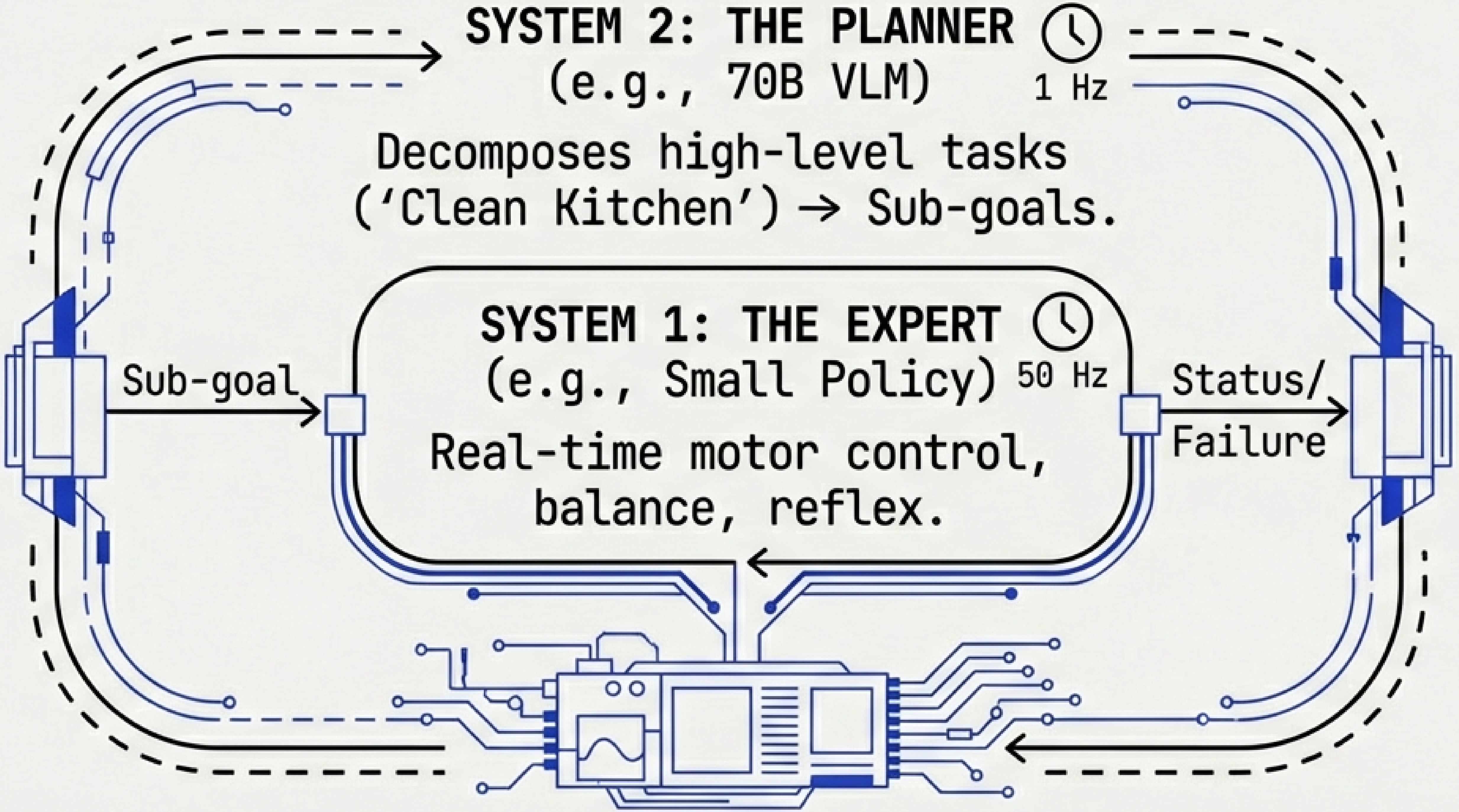


KNOWLEDGE INSULATION

Technique: Gradient blocking prevents robotic training from degrading general VLM internet knowledge.

DUAL-SYSTEM VLA ARCHITECTURES

DECOUPLING REASONING FROM REFLEX



PERFORMANCE

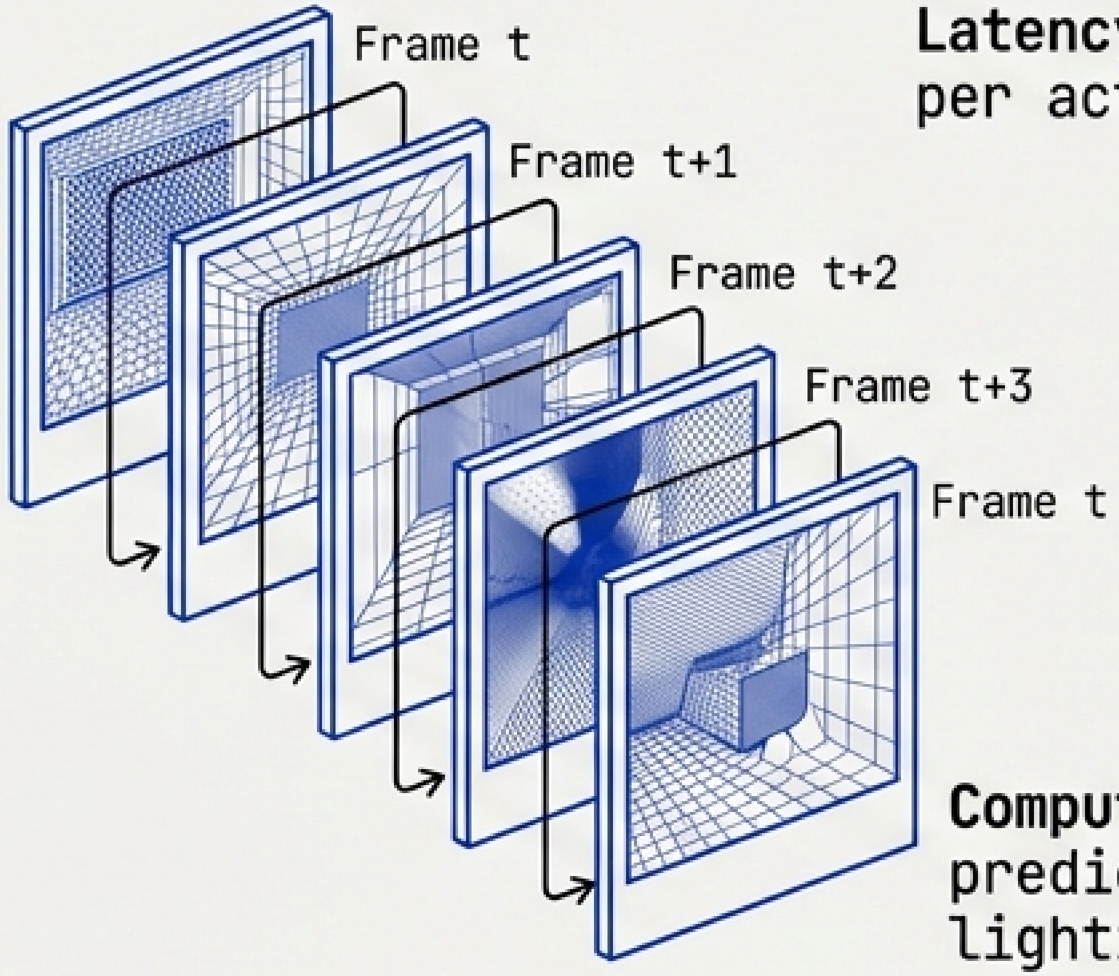
- 60-90% Real-World Success.
- Enables failure recovery (Slip → Replan).
- Reference Models: π0, Helix, GR00T.

WORLD MODELS: SIMULATION vs. LATENT PLANNING

OFFLINE TRAINING vs. ONLINE CONTROL

NVIDIA COSMOS	VL-JEPA
Pixel Rollout (Offline)	Latent Planning (Online)

NVIDIA COSMOS — Pixel Rollout (Offline)



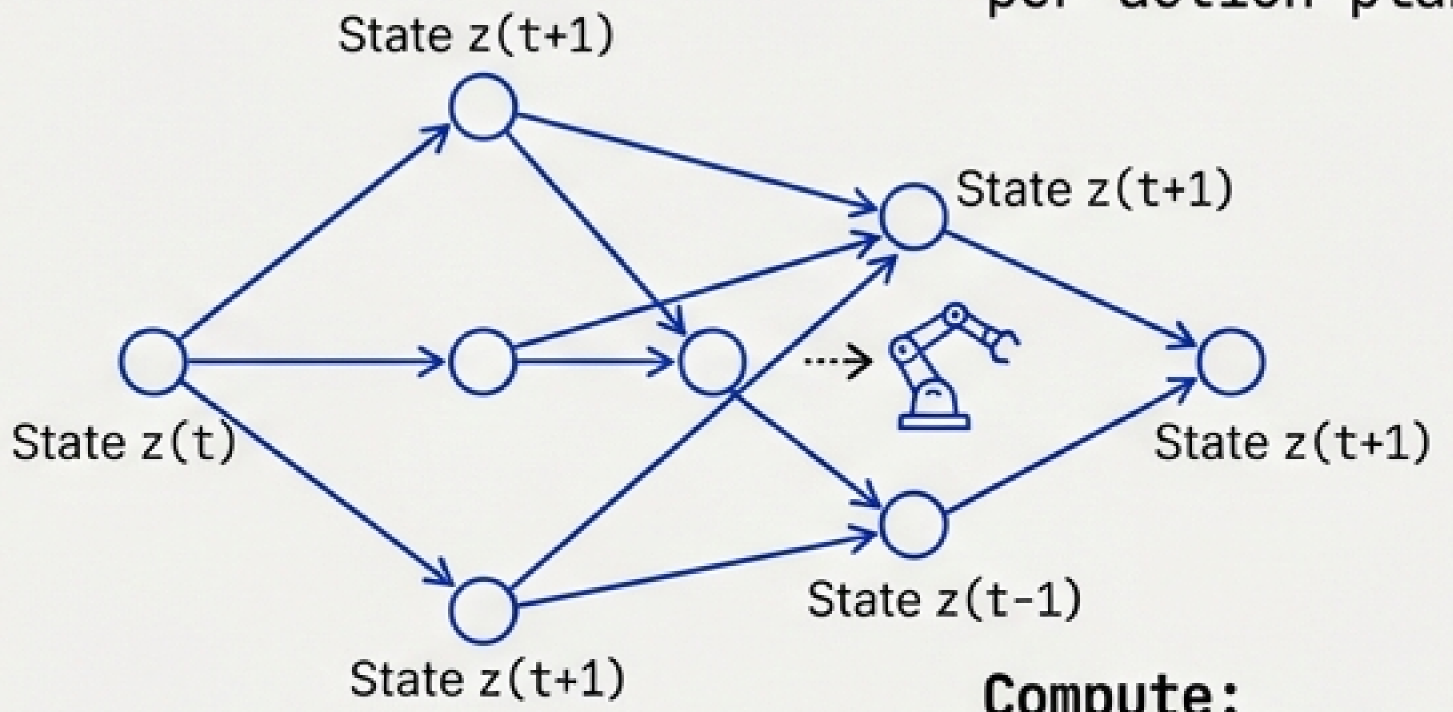
Latency: ~4 Minutes per action plan.

Compute: Must predict noise, lighting, texture.

Use Case: Synthetic Data Generation.

VL-JEPA — Latent Planning (Online)

Latency: ~16 Seconds per action plan.



Compute: Predicts only object physics/semantics.

Use Case: Real-Time MPC (Model Predictive Control).

RESULT: JEPA PROVIDES 15x SPEEDUP FOR ROBOTIC CONTROL.

ARCHITECTURAL SCORECARD: 2026 METRICS

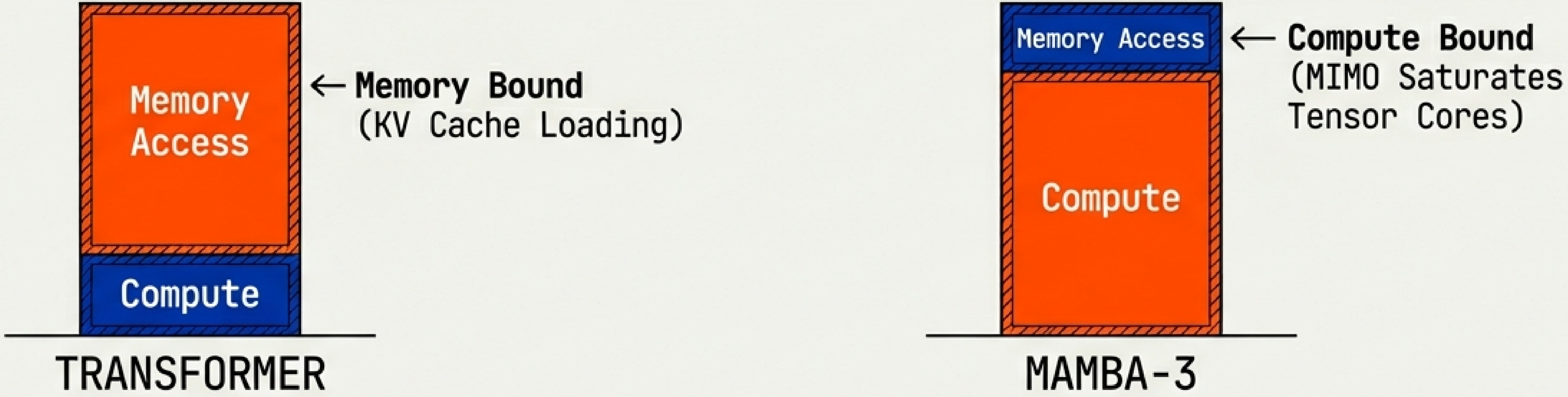
TECHNICAL COMPARISON MATRIX

FEATURE	TRANSFORMER (GPT-4)	SSM (MAMBA-3)	VL-JEPA
COMPLEXITY	Quadratic $O(n^2)$	Linear $O(n)$	N/A (Single Pass)
INFERENCE MODE	Autoregressive	Recurrent / Chunked	Predictive (Masked)
CONTEXT LIMIT	~128k (Soft)	100k+ (Hardware Limit)	Frame-based
STATE TRACKING	Perfect (Full History)	High (Complex State)	Semantic Only
PRIMARY USE	Reasoning / Coding	Long-Context / DNA	Real-Time Perception

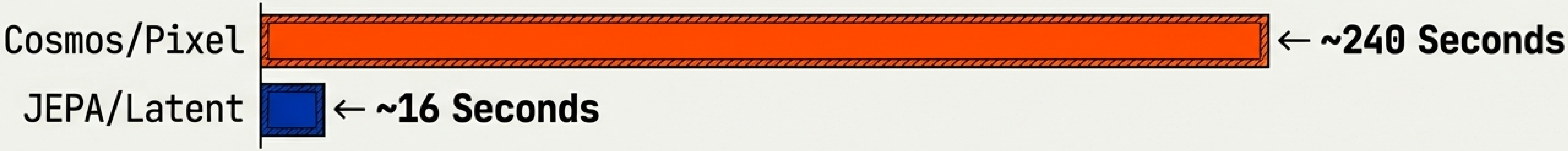
HARDWARE REALITY: LATENCY & ARITHMETIC INTENSITY

WHY ARCHITECTURE DICTATES DEPLOYMENT

MEMORY BOUND vs COMPUTE BOUND



PLANNING LATENCY COMPARISON



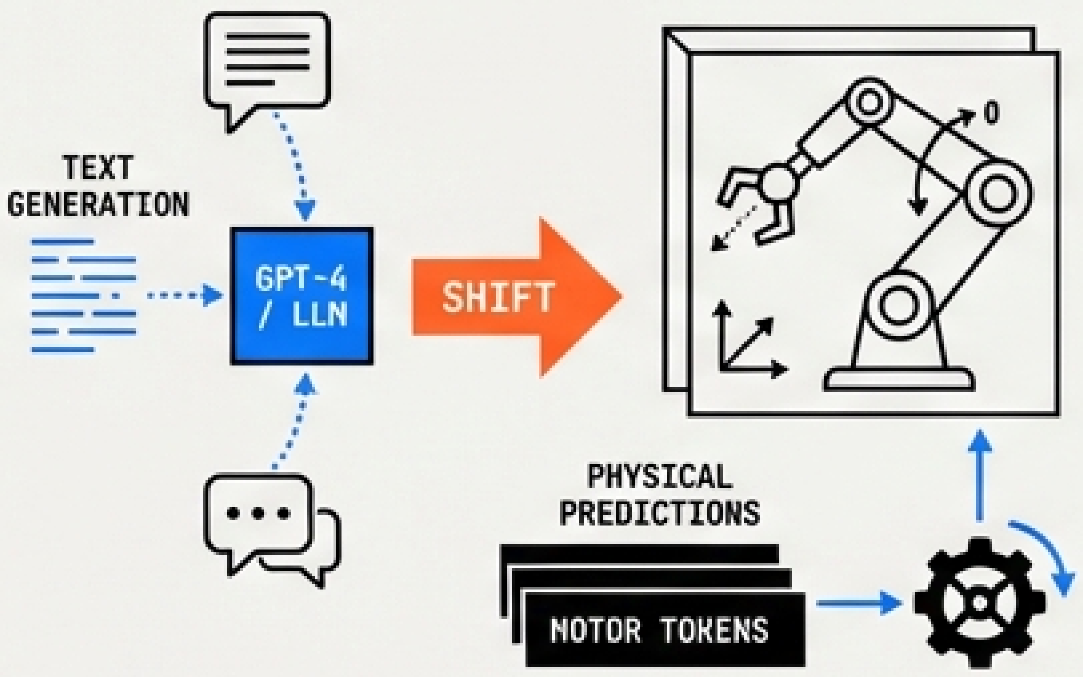
Streaming Efficiency: VL-JEPA ignores static frames for 2.85x fewer operations.

THE STRATEGIC SHIFT: FROM CHATBOTS TO PHYSICAL AGENTS

THE 2026 LANDSCAPE

OUTPUT MODALITY

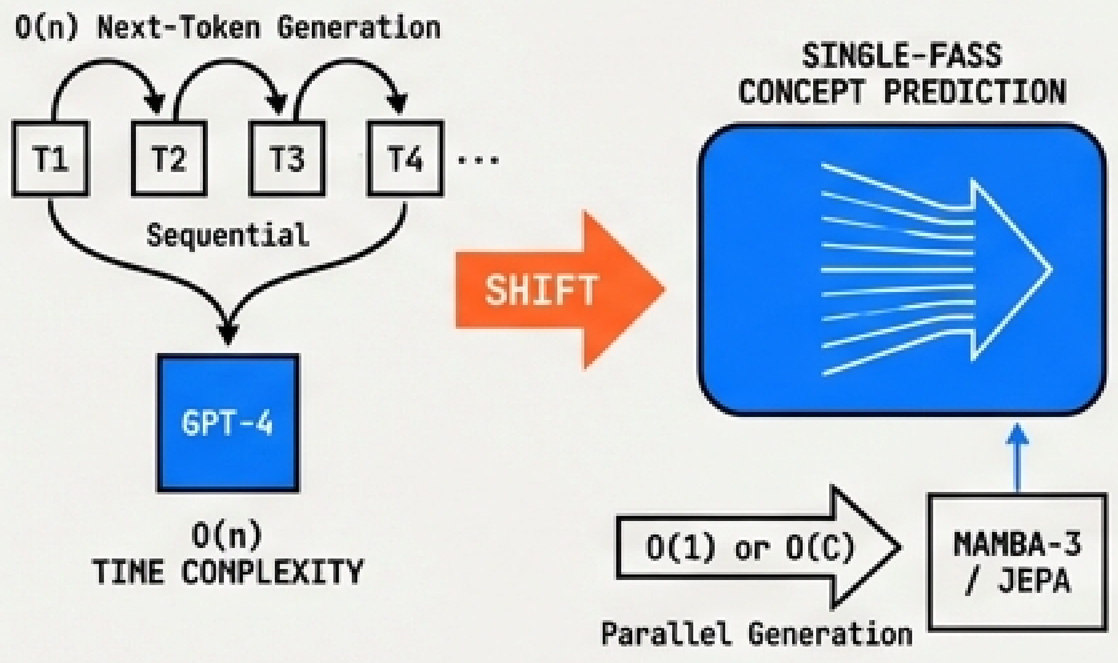
From **TEXT** → To **ACTIONS**



Shift: Generating physical predictions and motor tokens, not just language.

COMPUTATION PARADIGM

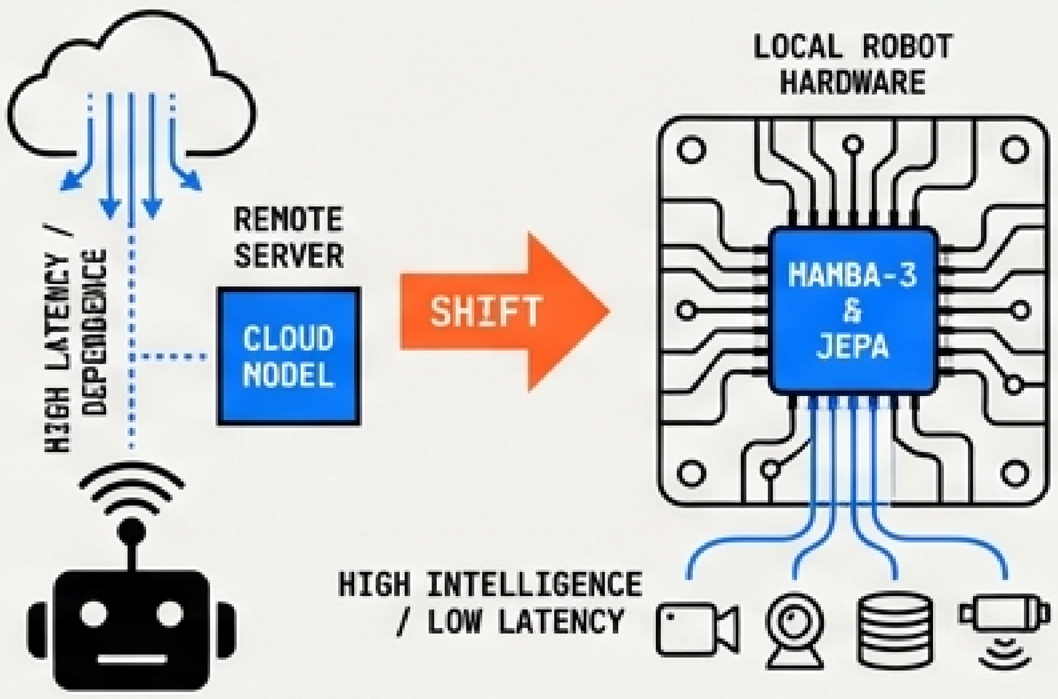
From **AUTOREGRESSION** → To **PREDICTION**



Shift: Moving from O(n) next-token generation to single-pass concept prediction.

EDGE INDEPENDENCE

From **CLOUD** → To **LOCAL**



Shift: Mamba-3 and JEPA enable high-intelligence models on local robot hardware.

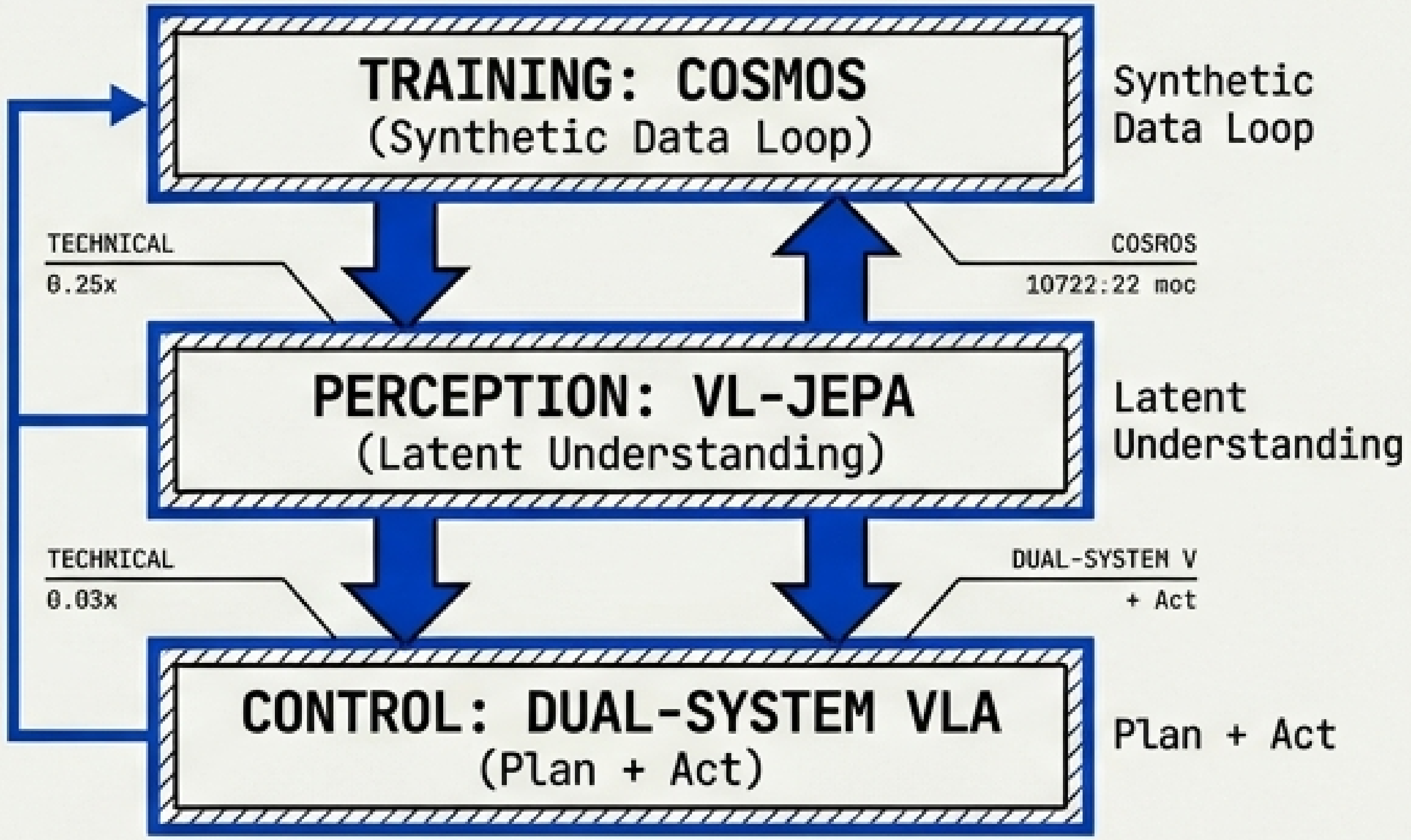
STRATEGIC TAKEAWAYS & THE WINNING STACK

DEPLOYMENT GUIDE

ENGINEERING RULES OF THUMB

- **For Reasoning: Stick to TRANSFORMERS (System 2).**
- **For Long Context:** Switch to MAMBA-3.
- **For Real-Time Vision:** Deploy VL-JEPA.
- **For Synthetic Data:** Utilize NVIDIA COSMOS.

THE 2026 HYBRID STACK



CONCLUSION: THE WINNING EMBODIED STACK IS BUILT ON LATENT EFFICIENCY FOR EXECUTION AND PIXEL SIMULATION FOR TRAINING.